Siyu Yuan

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RESEARCH INTEREST

Siyu Yuan（员司雨）is a third-year Ph.D. student at Fudan University in the School of Data Science. Her interested research topics are mostly around cognitive science with generative agents, including (but are not limited to):

- **Cognitive reasoning**, especially on exploring the cognitive reasoning abilities of generative agents, including analogical reasoning [3][4], concept understanding [6], and belief exploration of LLMs [10]. The ultimate goal is to enhance the understanding of these agents about themselves and others, thereby enabling them to generate responses that align better with human cognition.
- **Strategic Planning**, especially on equipping generative agents with human-level planning capabilities revolving around constrained planning [1] [5], tool invocation [2] [11] and other planning tasks.
- **Knowledge Acquisition**, especially on excavating knowledge based on generative agents, including concept acquisition [6] [7] [8], script generation [5], idiom construction [9] and analogy making [3][4]. These aim to construct rich knowledge resources that can be effectively utilized.

EDUCATION

Ph.D. in Statistics. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2021-2026 (estimated)*

B.S. in Data Science. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2017-2021*

SELECTED PUBLICATIONS

[*: Equal Contribution; ♠: Student I Mentored]

[1] *TIMEARENA: Shaping Efficient Multitasking Language Agents in a Time-Aware Simulation*
Yikai Zhang♠, **Siyu Yuan**, Caiyu Hu, Kyle Richardson, Yanghua Xiao, Jiangjie Chen
Long paper, Preprint, 2024.

[2] *EASYTOOL: Enhancing LLM-based Agents with Concise Tool Instruction*
Siyu Yuan*, Kaitao Song*, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan, Dongsheng Li, Deqing Yang
Long paper, Preprint, 2024.

[3] *AnalogyKB: Unlocking Analogical Reasoning of Language Models with A Million-scale Knowledge Base*
Siyu Yuan*, Jiangjie Chen*, Jiaqing Liang, Changzhi Sun, Yanghua Xiao, Deqing Yang
Long paper, Preprint, 2023.

[4] *Beneath Surface Similarity: Large Language Models Make Reasonable Scientific Analogies after Structure Abduction*
Siyu Yuan, Jiangjie Chen, Xuyang Ge♠, Yanghua Xiao, Deqing Yang
Long paper, in: **EMNLP 2023** (Findings).

[5] *Distilling Script Knowledge from Large Language Models for Constrained Language Planning*
Siyu Yuan, Jiangjie Chen, Ziquan Fu, Xuyang Ge♠, Soham Shah, Charles Robert Jankowski, Yanghua Xiao, Deqing Yang
Long paper, in: **ACL 2023**. **Outstanding Paper Award** (Top 1%)

[6] *Causality-aware Concept Extraction based on Knowledge-guided Prompting*
Siyu Yuan, Deqing Yang, Jinxi Liu, Shuyu Tian, Jiaqing Liang, Yanghua Xiao, Rui Xie
Long paper, in: **ACL 2023**.

[7] *Generative Entity Typing with Curriculum Learning*
Siyu Yuan, Deqing Yang, Jiaqing Liang, Zhixu Li, Jinxi Liu, Jingyue Huang, Yanghua Xiao
Long paper, in: **EMNLP 2022**.

[8] *Large-Scale Multi-granular Concept Extraction Based on Machine Reading Comprehension*
Siyu Yuan, Deqing Yang, Jiaqing Liang, Jilun Sun, Jingyue Huang, Kaiyan Cao, Yanghua Xiao, Rui Xie.
Long paper, in: **ISWC 2021**.

COMPLETE PUBLICATIONS

[9] *Deciphering Figurative Meanings: IdiomKB's Role in Optimizing Idiomatic Translation on Language Models*
Shuang Li♠, Jiangjie Chen, **Siyu Yuan**, Xinyi Wu, Hao Yang, Shimin Tao, Yanghua Xiao
Long paper, in: **AAAI 2024**.

[10] *Put Your Money Where Your Mouth Is: Evaluating Strategic Planning and Execution of LLM Agents in an Auction Arena*
Jiangjie Chen, **Siyu Yuan**, Rong Ye, Bodhisattwa Prasad Majumder, Kyle Richardson
Long paper, Preprint, 2023.

[11] *TaskBench: Benchmarking Large Language Models for Task Automation*
Yongliang Shen*, Kaitao Song*, Xu Tan, Wenqi Zhang, Kan Ren, **Siyu Yuan**, Weiming Lu, Dongsheng Li, Yueting Zhuang
Long paper, Preprint, 2023.

[12] *Contextual Information and Commonsense Based Prompt for Emotion Recognition in Conversation*
Jingjie Yi, Deqing Yang, **Siyu Yuan**, Kaiyan Cao, Zhiyao Zhang and Yanghua Xiao
Long paper, in: **ECML PKDD 2022**.

AWARDS AND ACHIEVEMENTS

ACL 2023 Outstanding Paper Award (Top 1%) *2023*
- Association of Computational Linguistics

Huatai Securities Scholarship *2023*
- Fudan University

Outstanding Graduate Student of Shanghai Colleges and University (Top 5%) *2021*
- Fudan University

Outstanding Student pacemaker of Fudan University *2021*
- Fudan University

China National Scholarship *2018*
- Fudan University

EXPERIENCE

Microsoft Research Asia (Shanghai) *September 2023 - Present*

- Role: Research Intern
- Advisor: Dr. Kaitao Song
- Topics: Autonomous Agents with Planning and Tool Use
- Project: Participate in the **JARVIS** project (Github Stars: 22.1K) [2] [11]

ByteDance AI-Lab (Shanghai) *January 2023 - May 2023*

- Role: Research Intern
- Advisor: Dr. Changzhi Sun and Dr. Jiaze Chen
- Topics: Evaluation and Improvement on the analogical reasoning abilities of LLMs [3].

Brain Technologies, NLP Research Intern (Remote) *June 2022 - September 2022*

- Role: Research Intern
- Advisor: Dr. Charles Jankowski
- Topics: Symbolic Knowledge Distillation for Language Planning and LLM Prompt Engineering [5].

Knowledge Works Lab at Fudan University (Shanghai) *March 2019 - Present*

- Role: Student Researcher
- Advisor: Prof. Yanghua Xiao and Dr. Jiangjie Chen
- Topics: Knowledge Acquisition, e.g, Analogy-making [4], Entity Typing [7], and Concept Extraction [8] [6].

Academic Service

- Program Committee/Reviewer of EMNLP (2022, 2023), APIN (2022, 2023), ACL(2023), ARR (2023, 2024), IEA/AIE (2023, 2024), ICLR (2024), COLM (2024)